SWAROOP CHAND J C

MACHINE LEARNING ENGINEER & INDEPENDENT RESEARCHER

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RESEARCH INTERESTS

Information retrieval; retrieval-augmented generation; LLM and retrieval evaluation; AI/ML systems; reproducible machine learning.

EDUCATION

Master of Engineering (M.Eng.), Artificial Intelligence and Machine Learning **2026**
Manipal School of Information Sciences, MAHE | CGPA: 9.08

Bachelor of Engineering (B.E.) **2023**
Sapthagiri College of Engineering, Bengaluru

RESEARCH

Independent research: chunking evaluation protocols in RAG **2026**

- Asked whether conclusions about RAG chunking strategies change when retrieval is compared at a fixed number of chunks versus a fixed word budget.
- Designed a multiverse analysis on 9,253 held-out queries from SQuAD 2.0, COVID-QA, CUAD and Qasper: 16 chunkers x 16 BM25 configurations x 2 selection protocols x 3 evidence-span metrics, extended to semantic breakpoint and clustering chunkers with BGE-M3 dense retrieval.
- Found that budget matching lowers the share of settings favouring 512- over 128-word chunks by 70.9 percentage points (95% CI [64.8, 81.9]) and raises support for the shorter semantic-breakpoint arm by 55.6 points (95% CI [50.8, 63.1]); the dense-retrieval hypothesis was not established.
- Wrote analysis plans before confirmatory runs; released code, stored results, deviation logs and validation tools with tests and CI; exploratory answer-quality check with a Qwen3-4B reader.

RESEARCH OUTPUTS

Swaroop Chand J C. Chunking Verdicts in Retrieval-Augmented Generation Depend on the Evaluation Protocol: A Multiverse Analysis. Manuscript prepared for arXiv submission, 2026.

Swaroop Chand J C. Chunking verdicts under count- and budget-matched retrieval: code, analysis plans and results (v0.1.1). Zenodo, 2026. DOI: 10.5281/zenodo.24053941

PROFESSIONAL EXPERIENCE

RESMED INDIA | Bengaluru, India **Jun 2025 - Present**

Associate Machine Learning Engineer - AI Platform & Agent Systems **May 2026 - Present**

- Implemented Agent Hub and AWS AgentCore-backed lifecycle flows for registering, discovering, updating, retiring, and routing independently deployed agents through a unified execution layer.
- Built a standalone FastAPI/ReAct web-research agent with SSE traces; diagnosed cluster-egress blocking and added a three-backend fallback that restored grounded, source-cited responses.
- Developed a BMAD-based multi-agent pilot for SDLC and incident RCA, and consolidated approximately 14 onboarding capabilities into a unified FastAPI backend.

Data Scientist - Enterprise RAG & Knowledge Platforms **Jun 2025 - May 2026**

- Built and hardened core components of a self-service enterprise RAG platform used to create 150+ assistants, including 2 customer-facing deployments and 10+ active internal bots.
- Engineered hybrid retrieval, semantic chunking, reranking, per-source Top-K controls, metadata filtering, and multimodal ingestion across enterprise sources.
- Reduced PDF onboarding time by 31-40% through bounded parallel processing and embedding-pipeline optimization.

WATTWORKER INC. | Machine Learning Research and Developer **2023 - 2024**

- Developed RNN, CNN, XGBoost, Random Forest, and H2O AutoML models for HVAC forecasting using sensor data from 3 UK/US commercial buildings; improved reported model performance from 78% to 94%.
- Built physics-informed thermodynamic models and supervised-learning models for EV battery remaining useful life and range prediction.

TECHNICAL SKILLS

Research: experimental design, analysis plans, multiverse analysis, bootstrap and cluster-robust inference, reproducibility | **IR:** BM25, dense retrieval, hybrid search, reranking, OpenSearch, pgvector | **ML/LLM:** PyTorch, TensorFlow, Transformers, RAG, LLM evaluation, LangGraph | **Engineering:** Python, NumPy, Pandas, FastAPI, Docker, Kubernetes, AWS, GitHub Actions